

VSEPR

Learning outcomes

By the end of this section you should be able to:

- Describe the concept of an electron domain.
- Predict the electron domain geometry and the molecular geometry for species with up to four electron domains.
- Predict the impact of non-bonding pairs and multiple bonds on bond angles.

Previously, you have drawn Lewis formulas to represent different molecules on paper, but does this two-dimensional representation really show what a molecule looks like? Have you ever wondered or tried to imagine what a molecule looks like in three-dimensional space?

Consider two molecules we have already encountered thus far, BeCl_2 and BCl_3 . As part of [section S2.2.1a](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43106/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43106/>) you learned how to represent these molecules using Lewis formulas (**Figure 1**).

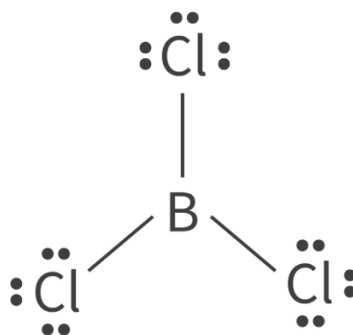


Figure 1a. Lewis formula BeCl_2

 More information for figure 1

The image depicts the Lewis structure of a BeCl_2 molecule, illustrating its molecular geometry. It shows a beryllium atom, represented as "Be," at the center connected by single lines (bonds) to two chlorine atoms, each denoted as "Cl." The chlorine atoms are positioned on either side of the beryllium. Each chlorine atom is surrounded by six dots representing three pairs of electrons, indicating the complete octet configuration. The structure visually emphasizes the bonds between the elements and the electron pairs surrounding the chlorines.

[Generated by AI]

Figure 1b. Lewis formula BCl_3

 More information for figure 1

The image is a Lewis structure diagram of BCl_3 , boron trichloride. The central atom is boron (B), which forms single bonds with three chlorine atoms (Cl). Each chlorine atom has three pairs of dots around it, representing the valence electrons. The diagram is structured in a trigonal planar geometry, with the boron at the center and the chlorine atoms evenly spaced around it, forming a triangle. No additional numerical values or labels are present in the diagram.

[Generated by AI]

Figure 1. Lewis formulas of BeCl_2 and BCl_3 .

Look at the central atoms in each one, surrounded by pairs of bonded electrons. What is the charge on these electrons? Because of this negative charge, what do you think would happen when two or three pairs of electrons are surrounding a central atom?

The electrons will repel each other. This means that the pairs of bonded electrons surrounding each central atom will orientate themselves so that there is maximum space between them.

Concept

Electrostatic forces: opposite charges are attracted to each other, similar charges repel. Much of our understanding of the structure of atoms, how they bond and how they form molecules is based on this.

The electrons around the central beryllium in BeCl_2 are found at 180° to each other. This is known as the **bond angle**; the angle between two bonds. What would be the bond angle for the electrons surrounding BCl_3 ? Look at **Figure 2**.

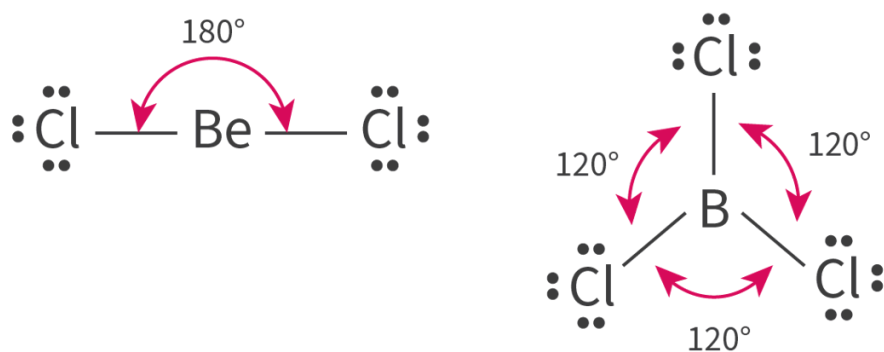


Figure 2. Bond angles for BeCl₂ and BCl₃.

 More information for figure 2

The image consists of two diagrams illustrating the bond angles in BeCl₂ and BCl₃ molecules. On the left, the BeCl₂ molecule is shown as a linear diagram with a beryllium atom at the center bonded to two chlorine atoms. The bond angle shown is 180 degrees, indicating a linear geometry. The electron pairs around the chlorine atoms are depicted as dots.

On the right, the BCl₃ molecule is structured in a triangular arrangement. A boron atom is centered, surrounded by three chlorine atoms. Each bond angle is marked as 120 degrees, demonstrating a trigonal planar geometry. The image uses arrows to emphasize the angles between the bonds, and the electron pairs are also represented as dots around each chlorine atom, illustrating the molecular geometry.

[Generated by AI]

The bond angles within a molecule are dependent on the number of pairs of electrons surrounding the central atom. These pairs of electrons are known as electron domains.

The electron domain geometry is dependent on how these electron domains are organised around a central atom. Each electron domain geometry has a specific name associated with it; for example, the electron domain geometry for BeCl₂ is linear and for BCl₃ it is trigonal planar (**Table 1**).

Table 1. Electron domain geometry.

Number of electron domains	Bond angle	Electron domain geomet
2	180°	Linear
3	120°	Trigonal planar

Let's now analyse another molecule, tetrachloromethane, CCl₄. Draw the Lewis formula for it and predict the bond angles between the pairs of electrons. Remember that you want to maximise the distance between the pairs of electrons

as they will repel each other (**Figure 3**).

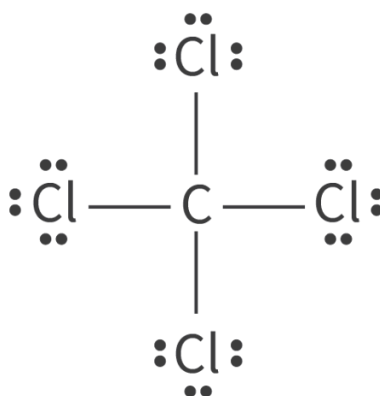


Figure 3. Lewis formula of tetrachloromethane, CCl₄.

 More information for figure 3

The image shows the Lewis structure for tetrachloromethane (CCl₄). In the center, there is a carbon (C) atom. Four chlorine (Cl) atoms are arranged symmetrically around the carbon, with single bonds connecting the carbon to each chlorine. Each chlorine atom has three pairs of dots representing lone pair electrons, arranged around them. The structure illustrates the tetrahedral geometry of the CCl₄ molecule.

[Generated by AI]

You may have guessed that the bond angle would be 90°. This would be the case if molecules existed in 2D. However, molecules exist in 3D and so they can exist on *x*, *y* and *z* planes. Explore this using the animation that follows. Click on the model in **Interactive 1** to rotate it and try to update your prediction of the bond angle.

Interactive 1. 3d Structure of Carbon Tetrachloride.

 More information for interactive 1

This interactive tool provides a three-dimensional visualization of carbon tetrachloride, allowing users to explore its structure, bonding, and molecular geometry. Users can rotate and manipulate the molecular representations to observe different perspectives. A carousel selector with arrows is at the bottom, allowing switching between a ball-and-stick model, a sphere model, and a structural formula. The carousel also includes options to highlight carbon, chlorine, and molecular bonds in the ball-and-stick model. In the ball-and-stick model, a small central sphere representing carbon is bonded to four larger spheres representing chlorine atoms. Chlorine atoms are connected to the carbon atom by thin rods, representing single bonds. This structure forms a tetrahedral geometry.

The sphere model depicts the same arrangement but without the connecting rods. In the structural formula, the central C atom is connected to Cl atoms at the top, right, bottom, and left via single bonds.

Below the selector are options labeled introduction, share, capture, augmented reality, and language.

The introduction option provides the following details:

Systematic name: Tetrachloromethane

Molecular formula: CCl_4

Molar mass: 153.82 g/mol

Description: Carbon tetrachloride is a colorless, toxic liquid with a mild, ether-like odor.

Applications: Carbon tetrachloride is an excellent solvent for non-polar compounds.

These three-dimensional representations allow users to explore carbon tetrachloride, including its bonding and geometry. They provide a clear view of the spatial arrangement of atoms in the molecule.

As you can see when you rotate the molecule in **Interactive 1** each bond is clearly greater than 90° . In fact, when there are four electron domains present, the molecular geometry will be tetrahedral and the bond angle will be 109.5° .

This linking of electron domains, bond angles and the resulting electron domain geometry is all part of an important theory known as the **Valence Shell Electron Pair Repulsion** theory, or VSEPR theory for short.

VSEPR and electron domain geometry

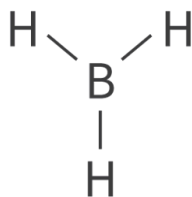
The **Valence Shell Electron Pair Repulsion** (VSEPR) theory is used to predict the electron domain geometry of the molecule. It is based on the ideas that:

1. Electron pairs, referred to as **electron domains** in VSEPR theory, repel each other. Note that it is focused only on the valence electrons of a central atom; electrons in inner-shells are not considered.
2. An electron domain can be **either** a single bond, a double bond, a triple bond or a non-bonded pair of electrons. Multiple bonds count as one domain.

Worked example 1

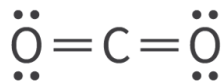
Identify the number of electron domains in a borane molecule (BH_3) and a carbon dioxide molecule.

Borane: BH_3



The central boron has no non-bonding pairs and three bonding pairs of electrons = 3 electron domains

Carbon dioxide: CO₂

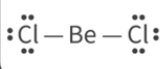


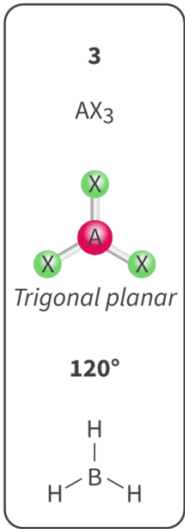
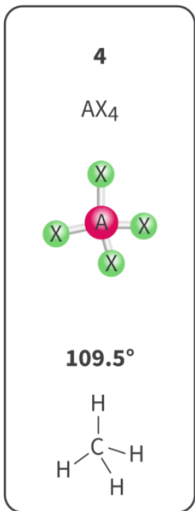
The central carbon atom has no non-bonding pairs of electrons and two double bonds = 2 electron domains

Once the number of electron domains are identified, we can use this to determine the bond angle and electron domain geometry of the molecule, i.e. where the electron domains are most likely to be found around the central atom. To do this, follow these steps:

1. Draw the Lewis formula.
2. Count the number of electron domains. Remember a double bond or a triple bond counts as one electron domain.
3. Determine the electron domain geometry and bond angles using **Table 2.**

Table 2. Determining electron domain geometry.

Number of electron domains	Bond angles	Electron domain geometry
2	Number of Electron Domains 2 Type AX_2 Electron Geometry  Bond Angle 180° Example  A = central atom, X = an atom or a group  More information The image is a diagram explaining the linear molecular geometry in chemistry. On the left, text describes various properties of this geometry: • 'Number of Electron Domains' is 2. • 'Type' is AX_2. • 'Electron Geometry' is Linear. • 'Bond Angle' is 180°. • 'Example' shows the molecule $Cl-Be-Cl$, with a central beryllium (Be) atom single-bonded to two chlorine (Cl) atoms in a straight line. The diagram also includes the label 'A = central atom, X = an atom or a group' at the bottom. [Generated by AI]	Linear

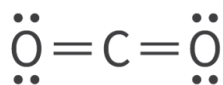
Number of electron domains	Bond angles	Electron domain geometry
3	3 AX_3  Trigonal planar 120° H $H-B-H$  More information This image is a diagram illustrating the trigonal planar molecular geometry. At the top of the image, the number '3' is displayed, indicating the molecular geometry type AX_3. Just below, a diagram shows a central atom labeled 'A' connected by lines to three surrounding atoms labeled 'X', forming an equilateral triangle. The label 'Trigonal planar' is written under the diagram, and the bond angle of 120° is noted below that. At the bottom, a structural formula represents Boron Hydride (BH_3), with a central 'B' atom bonded to three hydrogen atoms 'H', arranged in a plane forming a 'V' shape. This demonstrates the molecular structure using both geometric and chemical representations. [Generated by AI]	Trigonal planar
4	4 AX_4  109.5° H $H-C-H$ H	Tetrahedral

Number of electron domains	Bond angles	Electron domain geometry
	 More information The image is a diagram depicting a tetrahedral molecular geometry. It shows a central atom labeled 'A' surrounded by four peripheral atoms labeled 'X', forming a tetrahedral shape. The bond angle indicated between these atoms is 109.5 degrees. Below the diagram, there's a structural formula of methane (CH₄), illustrating carbon (C) in the center bonded to four hydrogen atoms (H). The image includes the text labels '4', 'AX₄', and '109.5°', representing the number of peripheral atoms, the type of molecular geometry, and the bond angle, respectively. [Generated by AI]	

Worked example 2

Identify the electron domain geometry for a carbon dioxide (CO₂), borane (BH₃) and methane (CH₄) molecule.

Carbon dioxide: CO₂

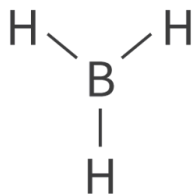


Number of electron domains around central atom: 2

Electron domain geometry: linear

Bond angle: 180°

Borane: BH₃

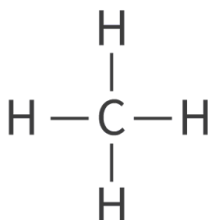


Number of electron domains around central atom: 3

Electron domain geometry: trigonal planar

Bond angle: 120°

Methane: CH_4



Number of electron domains around central atom: 4

Electron domain geometry: tetrahedral

Bond angle: 109.5°

Theory of Knowledge

You have learned how to represent molecular compounds by drawing a Lewis formula, a language consisting of accepted symbols and notations. Lewis formulas are two-dimensional drawings, yet molecules are three-dimensional objects. Think about a molecule of methane. A Lewis formula has the central carbon bonded to four hydrogen atoms, typically drawn at 90° angles, but the molecular geometry predicted by VSEPR theory states a tetrahedral formation with bond angles of 109.5° . How is your knowledge of a molecule's structure limited by the language used in a Lewis formula?

VSEPR and molecular geometry

Consider the Lewis formulas for NH_3 and NH_4^+ , which are drawn for you in **Figure 4**. What do you think the electron domain geometry of this molecule and ion would be? Do you think they would have the same bond angles?

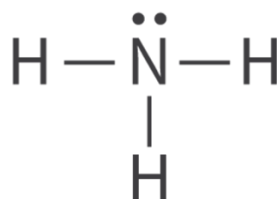


Figure 4a. Lewis formula NH_3

 More information for figure 4

The image is a Lewis structure diagram representing ammonia (NH_3). It features a nitrogen atom at the center symbolized by the letter 'N'. The nitrogen atom is bonded to three hydrogen atoms, each represented by the letter 'H'. The structure forms a trigonal pyramidal shape. There are single lines connecting the nitrogen atom to each hydrogen atom, indicating single covalent bonds. Additionally, there are two dots above the nitrogen atom representing a lone pair of electrons, which contribute to the molecule's pyramidal geometry.

[Generated by AI]

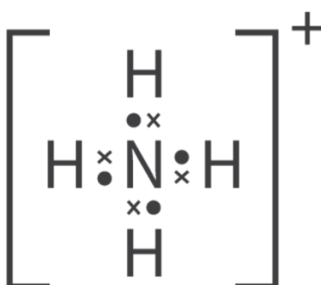


Figure 4b. Lewis formula NH_4^+

Figure 4. Lewis formulas of NH_3 and NH_4^+ .

Their electron domain geometry would both be tetrahedral as they both have four electron domains.

Bonded electrons and non-bonding pairs of electrons are repulsive in nature. Arranging these electron domains around the central atom in the shape of a cross maximises the distance between these electron pairs but it assumes their repulsive effects are all equal.

However, the non-bonding pairs of electrons are actually more repulsive than bonded pairs as they have a greater degree of freedom to move around in the electron cloud. This greater repulsion caused by the non-bonding pairs reduces the bond angles within molecules.

You can see this in the ball-and-stick models of NH_3 and NH_4^+ in **Figure 5**, where the non-bonding pair of electrons on NH_3 causes the bond angle to reduce from 109.5° to 107.3° .

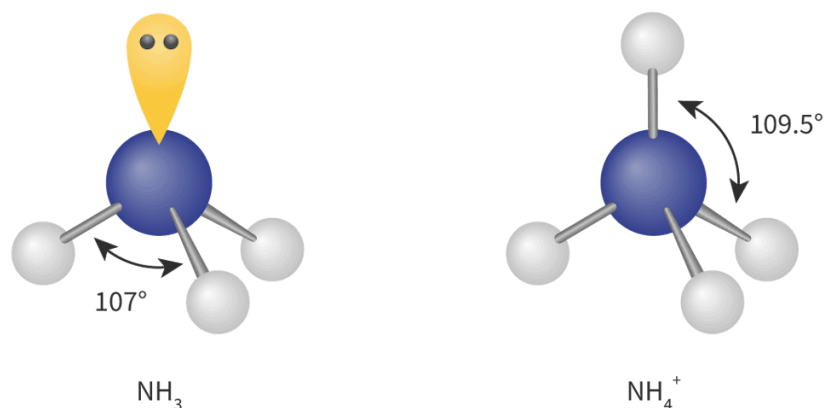


Figure 5. Ball and stick models of NH_3 and NH_4^+ .

 More information for figure 5

The image presents ball-and-stick models of the molecules NH_3 and NH_4^+ . Both are depicted with a central blue sphere representing nitrogen. In the NH_3 model at the left, there are three white spheres attached by sticks, indicating hydrogen atoms. A yellow shape with two black dots symbolizes a non-bonding electron pair on top of the central sphere. The bond angle is marked as 107° , showing the reduced angle due to electron pair repulsion. On the right, the NH_4^+ model lacks the non-bonding electron pair, showing four hydrogen atoms symmetrically arranged with bond angles of 109.5° each. This illustrates VSEPR theory's concept that non-bonding electron pairs cause more repulsion, thereby affecting molecular shape and bond angles.

[Generated by AI]

This is another key part of VSEPR theory: non-bonding pairs of electrons cause more repulsion than bonding pairs and the shapes and angles predicted should reflect this. The greatest repulsion occurs between two non-bonding pairs, followed by a non-bonding pair to a bonded pair and the least repulsion occurs between two bonded pairs of electrons.

Although electron domain geometry allows us to create a picture of where the electron clouds are located around a central atom, it does not indicate the arrangement of the atoms that are bonded to the central atom.

This is better described by the molecular geometry, which describes the arrangement of atoms, not electrons. Molecular geometry allows us to make an important distinction between a molecule such as NH_3 and a molecular ion such

as NH_4^+ .

Study skills

Electron domain geometry is based on the arrangement of electron pairs around a central atom, not the atoms. It incorporates bonded and non-bonding pairs of electrons.

Molecular geometry is focused on the atoms located around a central atom. It only incorporates bonded pairs of electrons.

The electron domain geometry and molecular geometry will be the same for molecules with no non-bonding pairs.

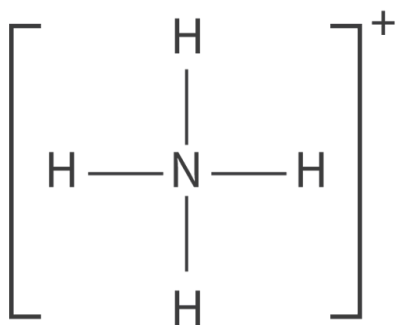
Both NH_3 and NH_4^+ have the same tetrahedral electron domain geometry since they both have four electron domains. The molecular geometry for NH_4^+ is also tetrahedral since the four electron domains are all bonding pairs, but for NH_3 , the molecular geometry is trigonal pyramidal since one of the electron domains is a non-bonding pair.

The molecular geometry will only be different from the electron domain geometry for central atoms that have non-bonding pairs of electrons.

Worked example 3

Identify the molecular geometry, including bond angle, for the ammonium ion and an ammonia molecule.

Ammonium: NH_4^+



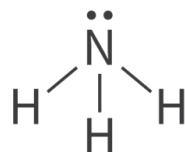
Number of electron domains around central atom: 4

Electron domain geometry: tetrahedral

Molecular geometry: tetrahedral

Bond angle: 109.5°

Ammonia: NH_3



Number of electron domains around central atom: 4 (one non-bonding pair and three bonded pairs).

Electron domain geometry: tetrahedral

Molecular geometry: trigonal pyramidal

Bond angle: 107° ($<109.5^\circ$)

What do you think would happen if two out of four electron domains were non-bonding pairs of electrons? How would it affect the bond angle?

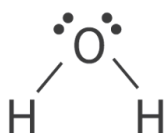
There would be an even greater decrease in the bond angle. The presence of two non-bonding pairs causes the bond angle to decrease to 104.5° and the molecular geometry to be bent.

You can see this in water molecules in **Worked example 4**.

Worked example 4

Identify the electron domain geometry, molecular geometry and bond angle, for a water molecule.

Water: H_2O



Number of electron domains around central atom: 4 (two non-bonding pair and two bonded pairs).

Electron domain geometry: tetrahedral

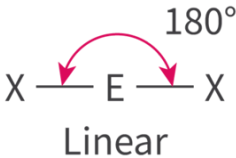
Molecular geometry: Bent

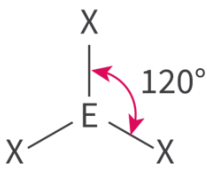
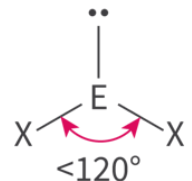
Bond angle: 104.5° ($<109.5^\circ$)

To determine the molecular geometry:

1. Draw the Lewis formula.
2. Count the number of electron domains. Remember a double bond or a triple bond counts as one electron domain.
3. From the electron domains identified in step 2, count the number of these that are non-bonding pairs.
4. Determine molecular geometry and bond angles as per **Table 3**.

Table 3. Molecular geometry.

Number of electron pairs	No lone pairs	One lone pair	
2	 Linear  More information The image is a diagram labeled "Linear" depicting a 180-degree angle. It consists of a straight line with three labeled points: X, E, and another X on the opposite side. Above the line, there's a red arc with arrows pointing downwards at each end, indicating the 180-degree angle between the two X points. The letter E is positioned in the middle of the line		

Number of electron pairs	No lone pairs	One lone pair
	segment, with lines extending horizontally to the two X points on either side. [Generated by AI]	
3	 Trigonal planar  More information The image is a diagram of a trigonal planar molecular geometry. It illustrates a central atom E surrounded by three identical atoms labeled X. The three X atoms are connected to the central E atom by straight lines, which represent bonds. The atoms are evenly spaced out to form a triangular planar shape. The diagram highlights a 120-degree angle between two adjacent bonds, indicated by a curved arrow. The structure and text emphasize the symmetrical arrangement typical for molecules with this geometry. [Generated by AI]	 Bent or angular  More information The image is a diagram labeled "Bent or angular." It depicts a structure with a central atom labeled 'E' connected to two other atoms labeled 'X'. The bonds between E and each X form an angle labeled as less than 120 degrees. Red arrows illustrate the bent or angular nature of the structure, indicating the angle measurement. Two dots are placed above the central part, possibly indicating lone pairs. [Generated by AI]
4		

Number of electron pairs	No lone pairs	One lone pair
	Tetrahedral More information The image depicts a diagram of tetrahedral molecular geometry. Four 'X' symbols represent atoms surrounding a central 'E', indicating an element. The diagram includes a bond angle of 109.5 degrees between two of the 'X' atoms as depicted by red arrows. The 'X' symbols are positioned in a way that represents three-dimensional spatial orientation, with one 'X' appearing behind the plane (dashed lines) and another projecting out (bold triangle). The label 'Tetrahedral' is placed at the bottom. [Generated by AI]	Trigonal pyramidal More information The image shows a diagram depicting the trigonal pyramidal molecular geometry. At the center is an atom labeled 'E' with three atoms labeled 'X' connected to it, forming a pyramid-like structure. The central atom 'E' has a pair of lone electrons at the top represented by two dots. Beneath the central atom are three 'X' atoms positioned to form angles less than 109.5 degrees, with one of the bonds shown as a filled black triangle to indicate spatial orientation. Two red arrows are used to highlight the angle between the bonds. The diagram includes text indicating the angle is less than 109.5 degrees and labels the structure as 'Trigonal pyramidal.' Other lines represent the bonds with the central atom, showing one bond going into the plane of the screen and another coming out. This diagram illustrates the molecular shape where the central atom has three bonds and one lone pair. [Generated by AI]

Study skills

The molecular geometry associated with three electron domains is trigonal planar.

4 electron domains, one of which is a non-bonding pair, results in a trigonal pyramid geometry.

For both of these geometries it is also acceptable to say triangular instead of trigonal, e.g. triangular planar or triangular pyramid. The same thing arises with bent geometry — this can also be known as 'angular' or 'v-shaped'.

In both situations, just pick one and stick to it so you do not get confused!

Previously in this section, we identified the electron domain geometry of a borane molecule, BH_3 , as trigonal planar due to its three electron domains and its bond angle as 120° . As it has no non-bonding pairs, what is its molecular geometry?

How would the bond angle be affected if one of the three electron domains was a non-bonding pair?

Worked example 5

Identify the electron domain geometry, molecular geometry and bond angle, for the molecule of XY_2 shown:

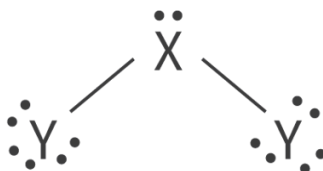


Figure 6. Identify the electron domain geometry, molecular geometry and bond angle.

 More information for figure 6

The diagram illustrates a molecule with a central atom, labeled X, which is connected to two surrounding atoms, each labeled Y. The central atom X is shown with two pairs of electrons represented by two dots above the X. Each Y atom is also accompanied by electron pairs, depicted as three dots in a triangular arrangement around the letter Y. The angled arrangement of atoms creates a bent molecular shape, suggesting the electron domain and the molecular geometry, which appears similar to a bent or V-shaped molecule.

[Generated by AI]

Number of electron domains around central atom: 3 (one non-bonding pair and two bonded pairs).

Electron domain geometry: trigonal planar

Molecular geometry: Bent

Bond angle: 117° ($<120^\circ$)

Study skills

You may wonder why specific values are not used for some of the bond angles above.

There are several factors (besides the presence of non-bonding pairs) that can affect the specific bond angle for any molecule and so usually a range of values will be accepted in an exam.

When quoting bond angle values for molecular geometries, the symbol $<$ is used to indicate less than.

When you are identifying the bond angle associated with a specific molecular geometry, estimate a value and in brackets include

Electron Domain Geometry angle.

A good rule of thumb is that the bond angle will decrease by about 2° for each non-bonding pair of electrons around the central atom for 4 electron domains. For example, if the tetrahedral molecular geometry has an angle of 109.5° , trigonal pyramidal could be expected to be $\sim 107^\circ$ ($<109.5^\circ$) and bent $\sim 104.5^\circ$ ($<109.5^\circ$)

VSEPR and covalent network structures

Recall the structure of diamond from [section S2.2.0](#)

(<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43109/>) shown in **Figure 7**. Each sphere represents one carbon atom.

Pick one carbon and identify how many other carbons it is bonded to. Now do the same for another carbon. Do you think this will be consistent throughout the entire structure?

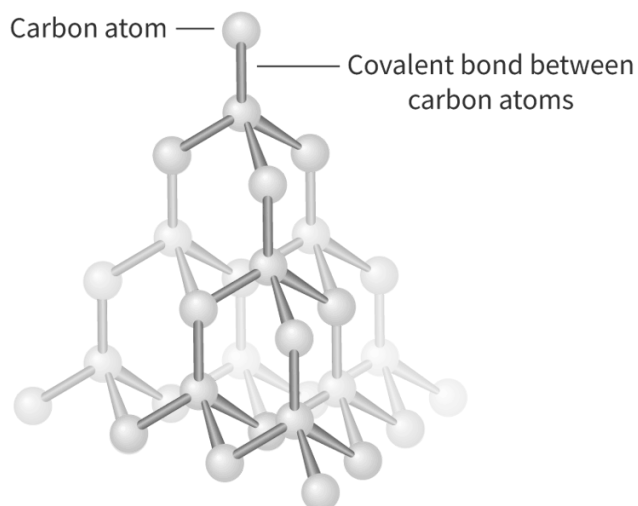


Figure 7. The structure of diamond in 3D.

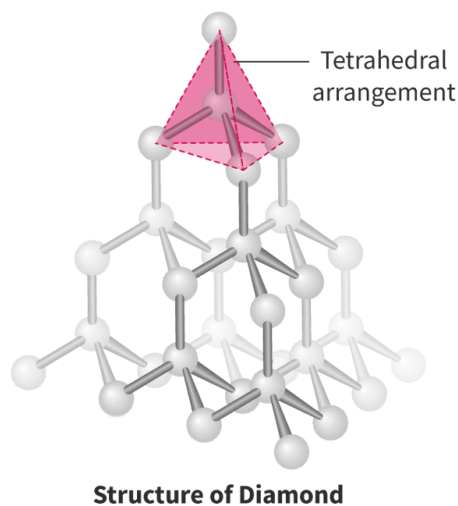
 More information for figure 7

The image depicts a 3D structure of diamond, illustrating how carbon atoms are bonded together in a crystalline lattice. Each sphere represents a carbon atom, and the lines connecting the spheres represent covalent bonds between them. The structure shows a repeating pattern where each carbon atom is bonded to four other carbon atoms in a tetrahedral arrangement. This creates a robust, interlocking structure characteristic of diamond. The repetitive bonding pattern is consistent throughout the crystalline framework.

[Generated by AI]

Remember that each covalent bond between the carbon atoms in diamond contains a pair of electrons that are negatively charged and will orient themselves so they are as far apart as possible. Using VSEPR theory, what electron domain and molecular geometries do you think would be found throughout this molecule?

They will orient themselves to have a tetrahedral geometry for both molecular and electron domains.



Diamond is an example of a covalent network structure. You will learn more about these structures in section S2.2.7 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/covalent-network-structures-id-43593/>).

For larger molecules, including macromolecular structures such as diamond with no one distinct central atom, it is important to treat each carbon as its own central atom. The same large macromolecule could have several different VSEPR geometries in it, so it is important to identify the bond angles for each one separately.

Worked example 6

Identify the molecular geometry, including bond angle, within the propyne molecule.

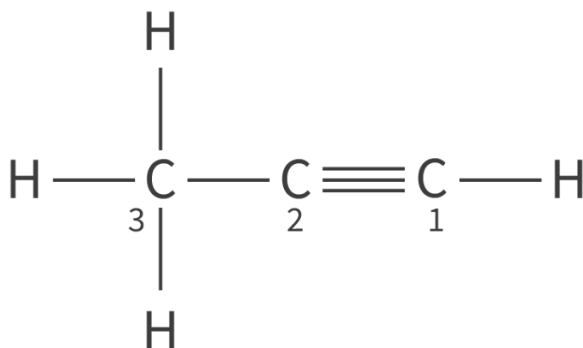


Figure 8. Identify the molecular geometry.

More information for figure 8

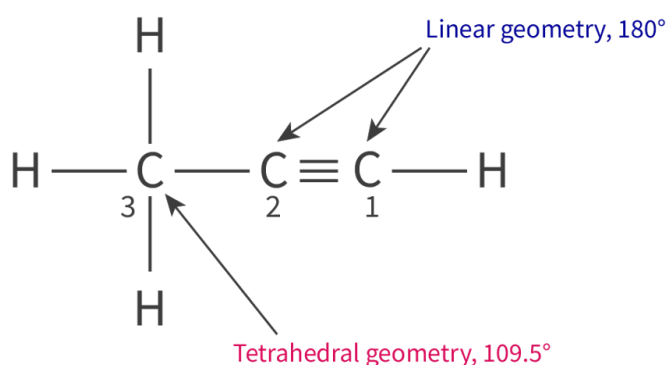
The image displays a structural diagram of a propyne molecule. It shows three carbon (C) atoms connected in a linear arrangement. The first carbon atom (C1) on the right is triple-bonded to the second carbon atom (C2), indicated by three parallel lines. The

third carbon atom (C3) is single-bonded to the second carbon atom, shown by a single line.

C1 has an additional bond extending to a hydrogen atom (H) on the right. C2 does not have any other bonds. C3 is connected to three hydrogen atoms, one above, one below, and one to the left, creating a trigonal planar geometry around it, consistent with sp^2 hybridization. The bond angles are indicated between each connected atom. Numbers (C1, C2, C3) are used to label the carbon atoms, while a subscript '3' is placed next to the hydrogen group bonded to C3, denoting three hydrogen atoms bonded to C3.

[Generated by AI]

Analyse each central atom separately to determine their individual molecular geometries.



C1:

Electron domains: 2 electron domains and no non-bonding pairs

Molecular geometry: linear

Bond angle: 180°

C2:

Electron domains: 2 electron domains and no non-bonding pairs

Molecular geometry: linear

Bond angle: 180°

C3:

Electron domains: 4 electron domains and no non-bonding pairs

Molecular geometry: tetrahedral

Bond angle: 109.5°

Summary

The relevant postulates of VSEPR are:

- Pairs of electrons, known as electron domains, in the valence shell repel each other due to their negative charge.
- Electron domains orient themselves around a central atom to reduce the repulsion between them by maximising the distance.
- When there is a double or triple bond between two atoms, treat this as one electron domain.
- The number of valence shell electron domains (bonded and non-bonded pairs) around the central atom determines the electron domain geometry of the molecule.
- The molecular geometry can be determined by factoring in the additional repulsion non-bonding pairs create compared to bonded pairs. If no non-bonding pairs are present, then the molecular geometry will be the same as the electron domain geometry.

Nature of Science

Aspect: Models

The VSEPR theory is useful in helping us deduce the three-dimensional geometries of molecules, explain properties and predict reactions. However, it is not flawless: there are times when its predictions do not hold true for particular irregularities or when the size of the substituents has an impact. If you are studying the Higher level, you will encounter this in the study of transition metal complexes. When models such as VSEPR prove inadequate, more sophisticated models must be developed.

Try this next activity to model different molecules on a macro scale.

Activity

- **IB learner profile attribute:** Thinker

- **Approaches to learning:** Thinking skills — Designing procedures and models
- **Time required to complete activity:** 30 minutes
- **Activity type:** Group activity

Representing electron domain and molecular geometry with balloons

Materials: Pack of 20 balloons, string or thread

In this activity, each balloon represents an electron domain. Tying or bonding each electron domain together with string will force balloons to occupy the position which gives them the most space relative to the other balloons, similar to how electron domains do this in real life.

Follow the instructions below to create the different geometries, and understand the molecular geometries that arise as a result of the additional repulsion caused by the presence of non-bonding pair(s).

Procedure 1: Electron Domain Geometry

A. Linear geometry

1. Inflate two balloons to the same size.
2. Tie the two balloons together as shown in **Figure 9**.

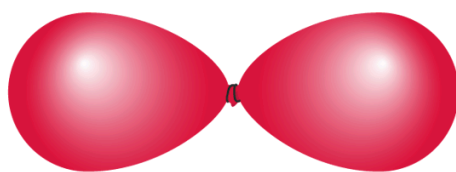


Figure 9. Linear geometry.

B. Trigonal planar geometry

1. Inflate three balloons to the same size.
2. Tie the three balloons together as shown in **Figure 10**.

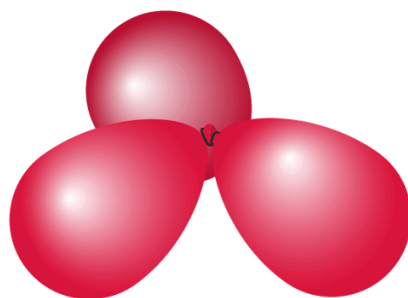


Figure 10. Trigonal planar geometry.

© More information for figure 10

C. Tetrahedral geometry

1. Inflate four balloons to the same size.
2. Tie the balloons into pairs as shown in **Figure 9**.
3. Connect the two pairs together by twisting the 'bond' point of one pair around the 'bond' point of another as per **Figure 11**.

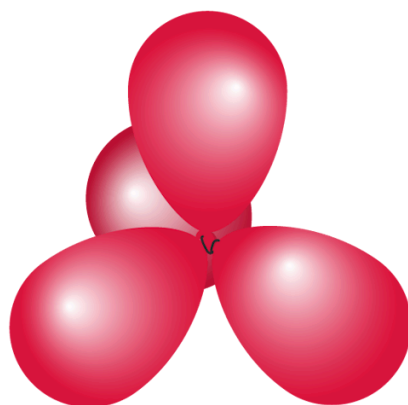


Figure 11. Tetrahedral geometry.

👁 More information for figure 11

Procedure 2: Molecular Geometry

D. Bent / angular geometry

1. Inflate two balloons to the same size and one balloon to be larger.
2. Tie the three balloons together as shown in **Figure 12**.
3. Compare the positioning of the balloons and angles between them to what was produced in Procedure 1B.

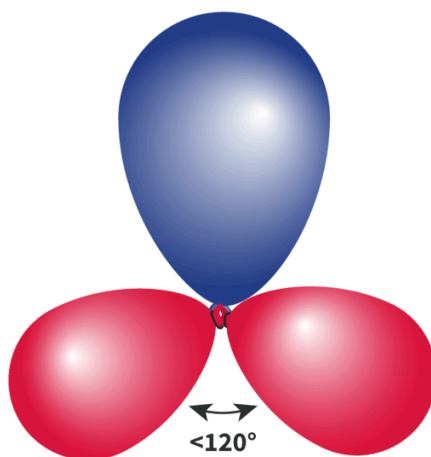


Figure 12. Bent/angular geometry.

👁 More information for figure 12

E. Trigonal pyramid

1. Inflate three balloons to the same size and one balloon to be larger.
2. Tie the four balloons together as shown in **Figure 13**.
3. Compare the positioning of the balloons and angles between them to what was produced in Procedure 1C.

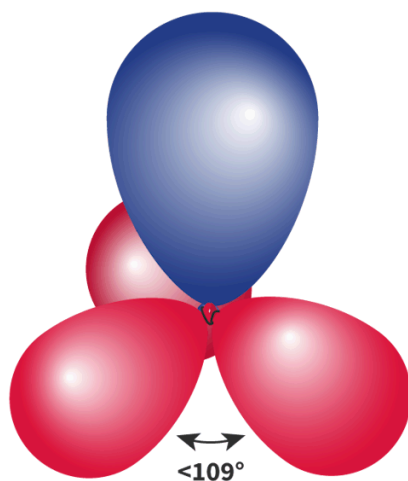


Figure 13. Trigonal pyramidal geometry.

📄 More information for figure 13

F. Bent / angular geometry

1. Inflate two balloons to the same size and two balloons to be larger.
2. Tie the four balloons together as shown in **Figure 14**.
3. Compare the positioning of the balloons and angles between them to what was produced in Procedure 1C and 2E.

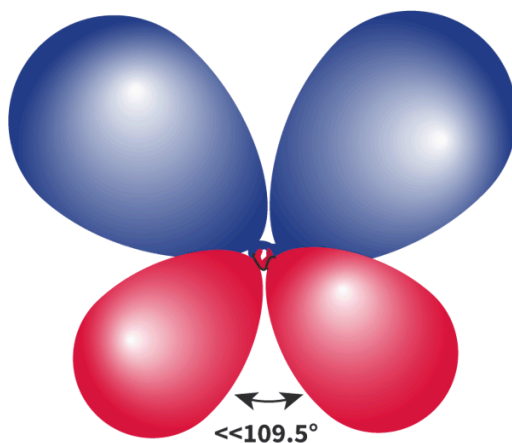


Figure 14. Bent/angular geometry.

📄 More information for figure 14

Reflection: Consider what you have learned by modelling the VSEPR geometries using the balloons. Are balloons an effective way to replicate the geometries? Suggest another material that could also be used to demonstrate the electron domains.

